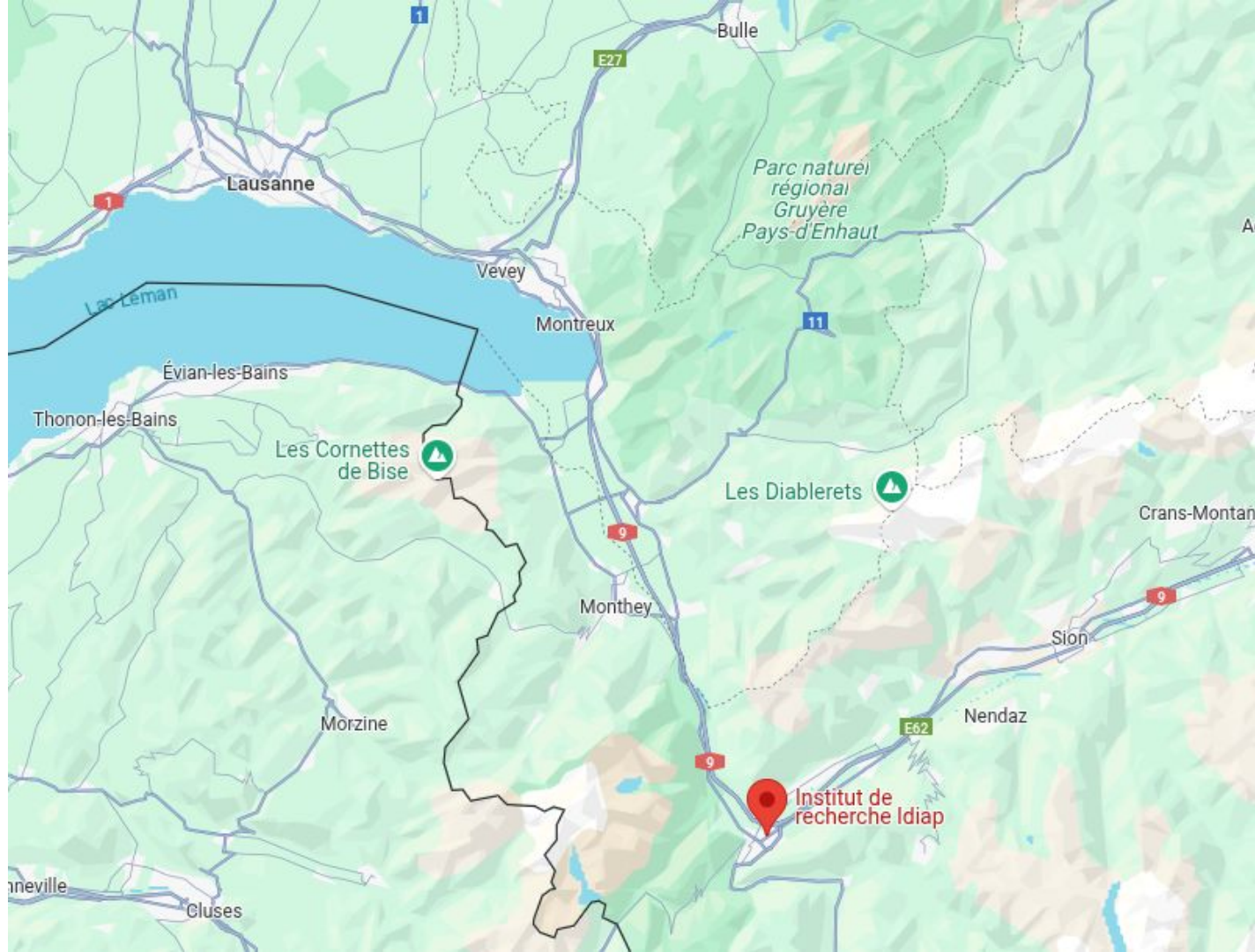


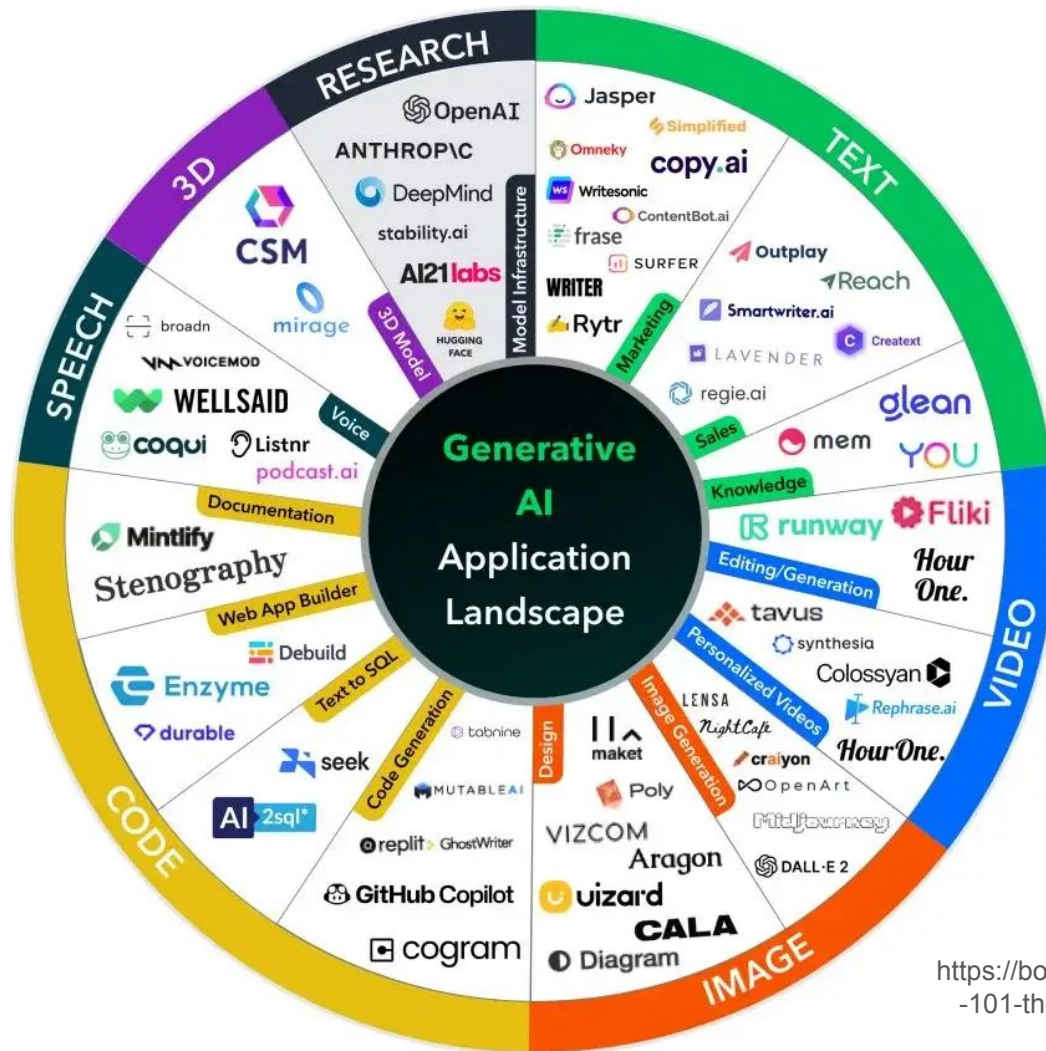
Generative models

Presented by Alina (Mingchi Hou)

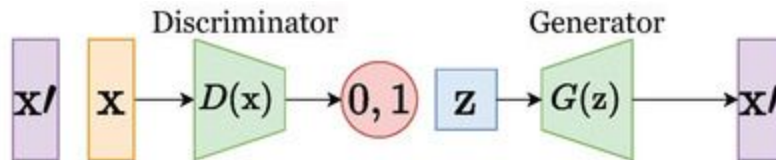
Lauzhack

July 16, 2025

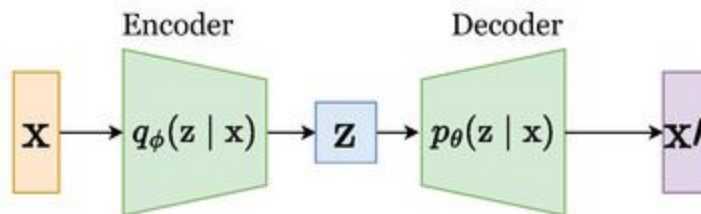




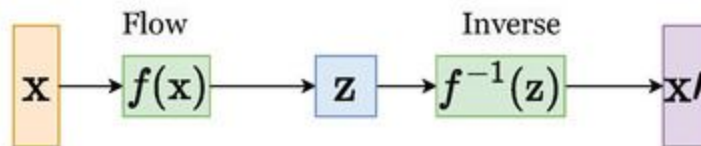
GAN:
adversarial
training



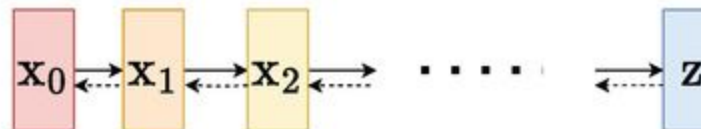
VAE: maximize
variational
lower bound



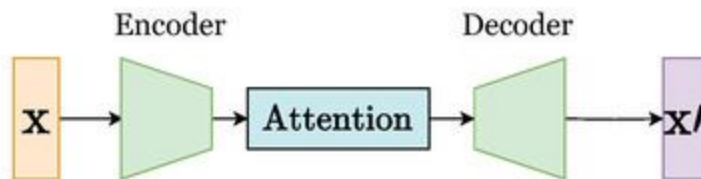
Flow-based models:
invertible transform of
distributions



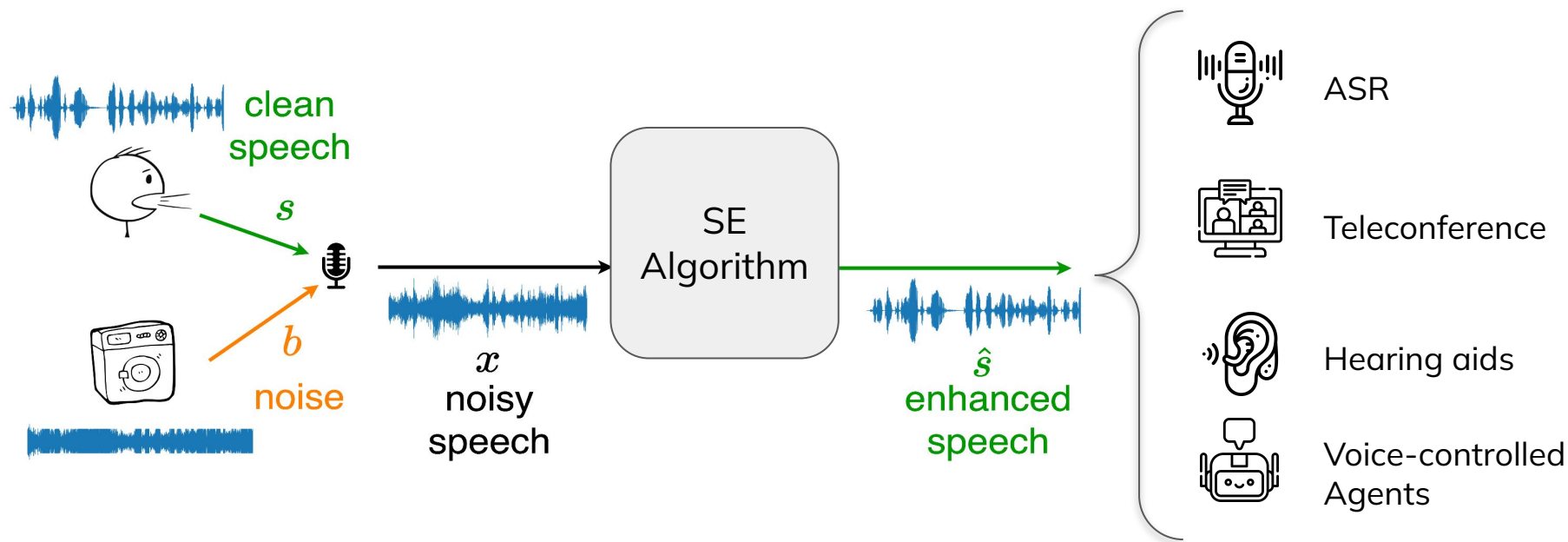
Diffusion models:
gradually add Gaussian
noise and then reverse



Attention mechanism:
autoregressive series
completion

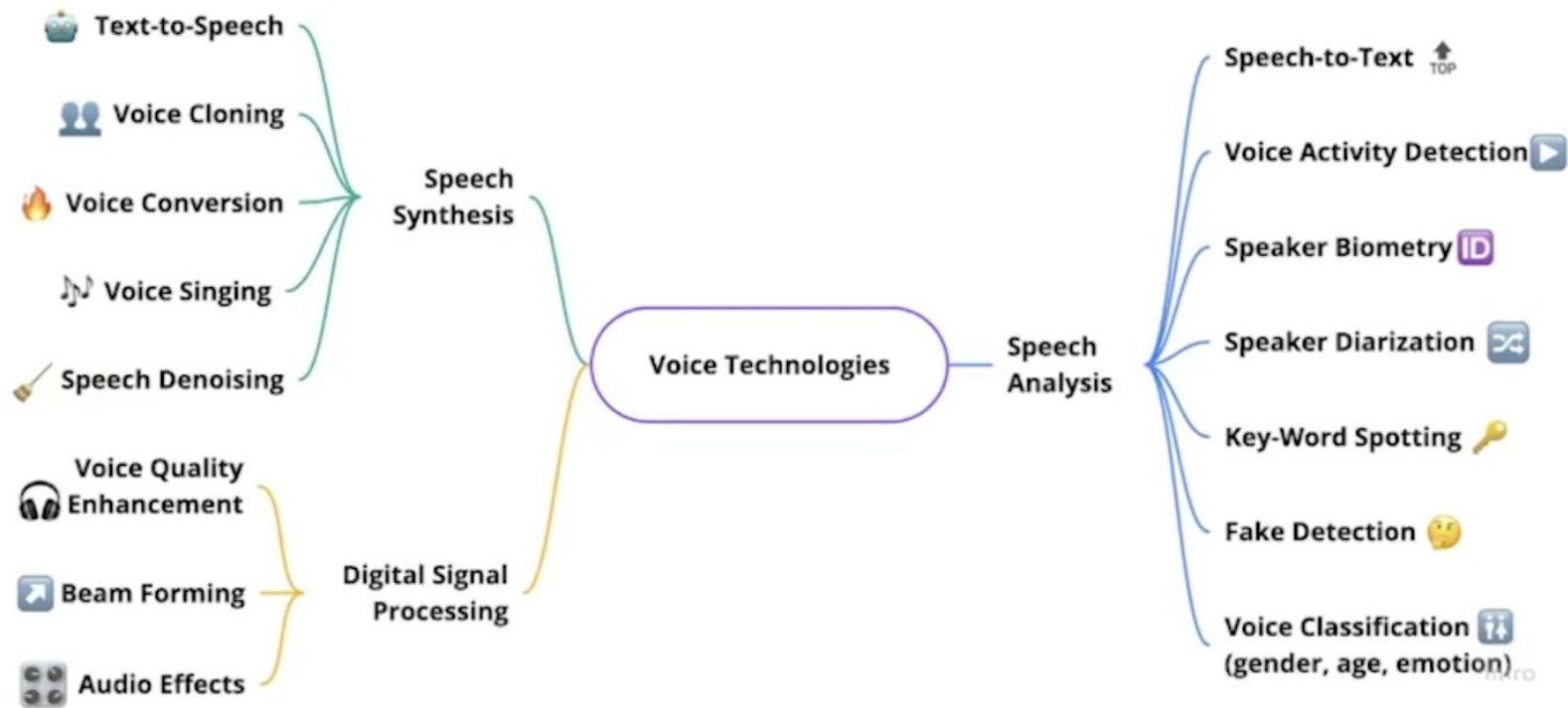


Speech Enhancement

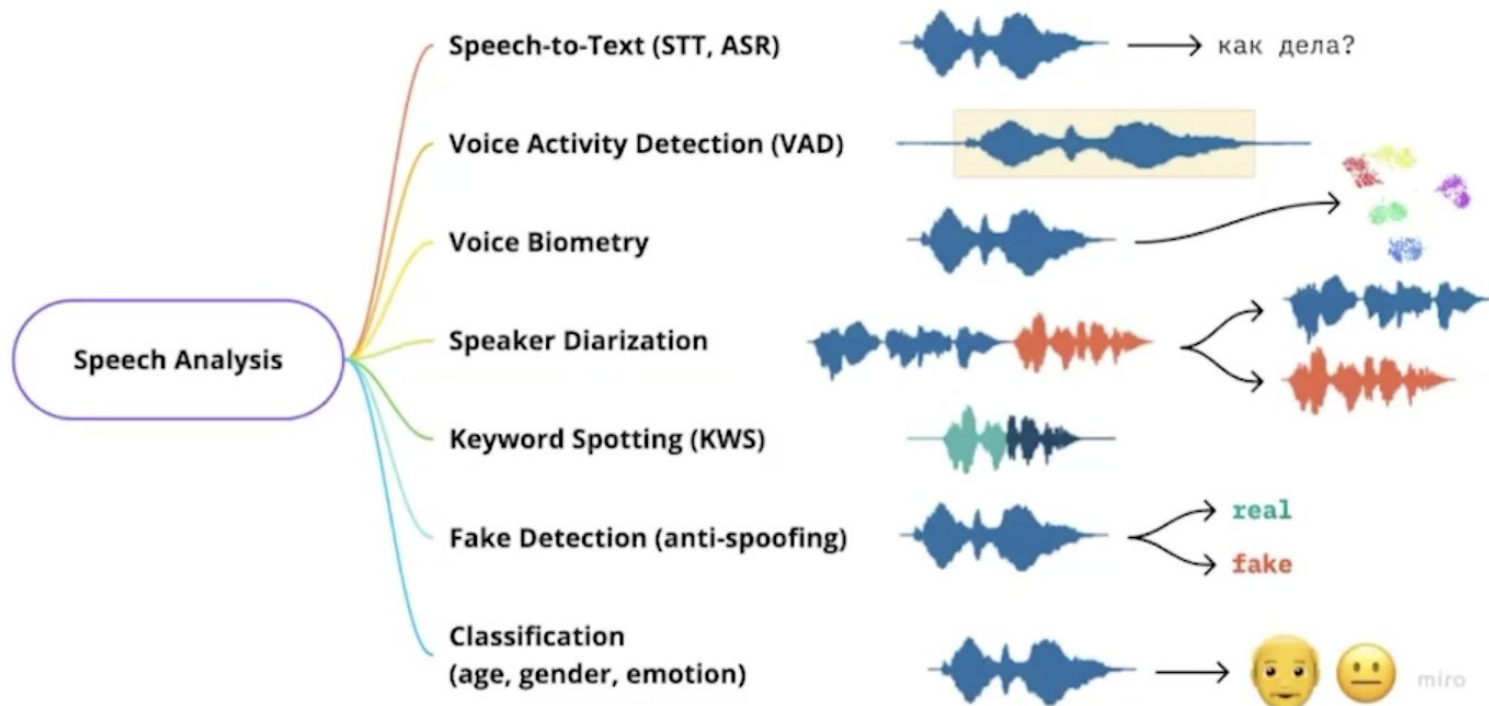


(Adapted from Signal Processing Group, Universität Hamburg)

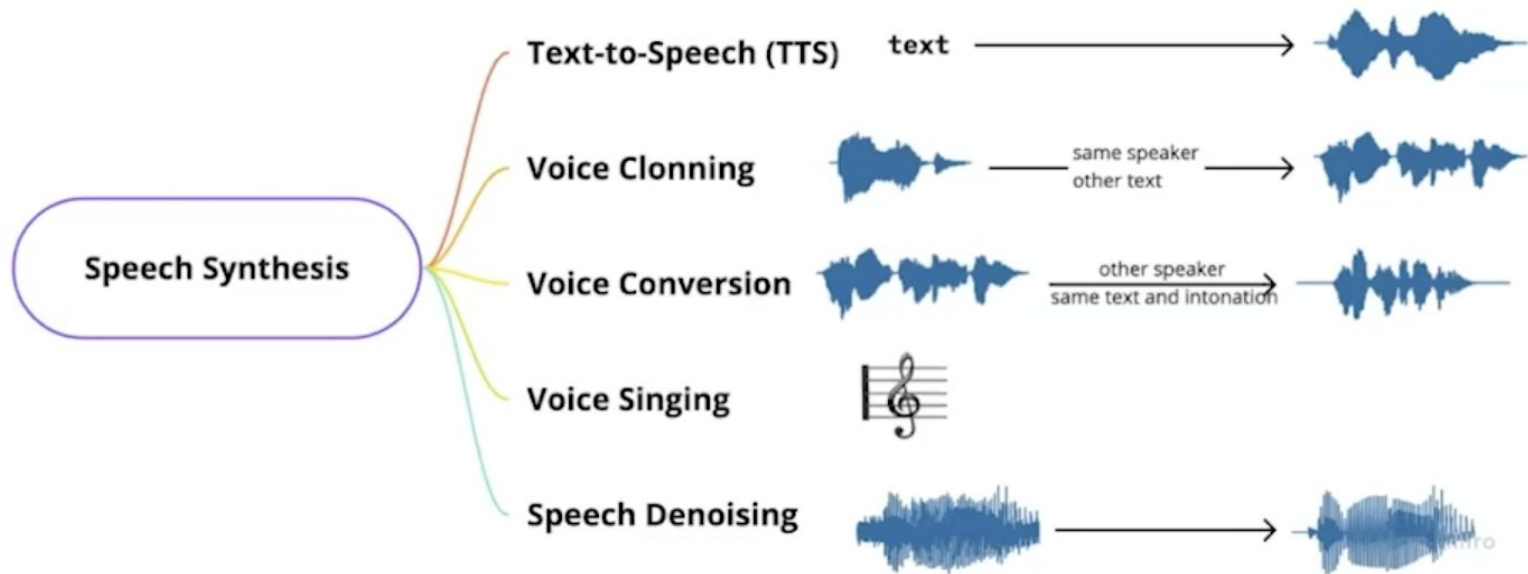
Voice Technologies Mind Map



Voice Technologies Mind Map



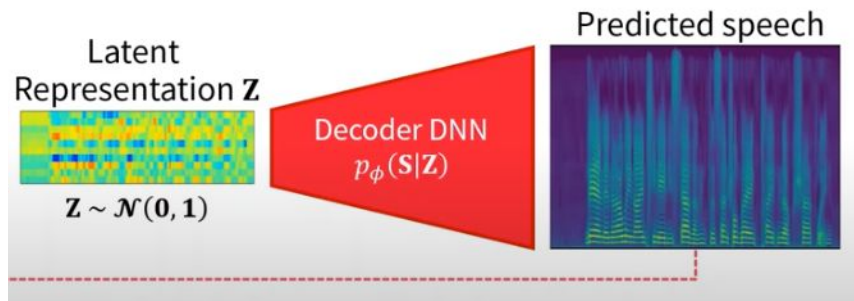
Voice Technologies Mind Map



VAE

Variational Autoencoder

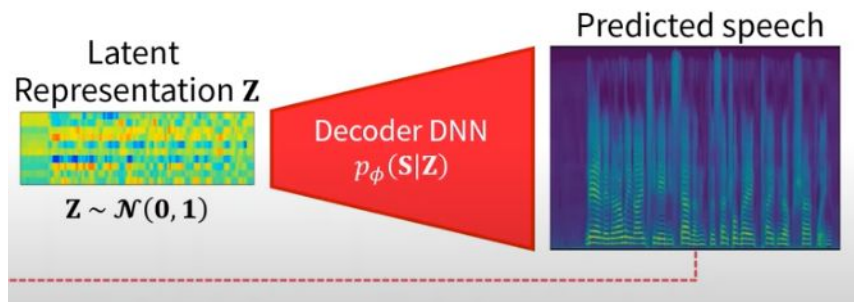
Variational AutoEncoder



- Prior $p_{\theta}(\mathbf{z})$
- Likelihood $p_{\theta}(\mathbf{x}|\mathbf{z})$
- Posterior $p_{\theta}(\mathbf{z}|\mathbf{x})$

(Adapted from Bando et, al. Interspeech 2020)

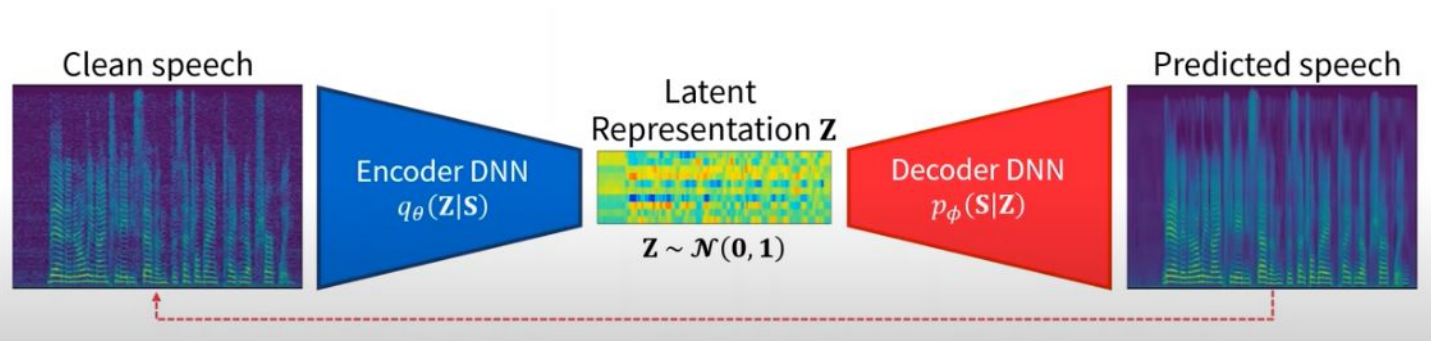
Variational AutoEncoder



- Prior $p_{\theta}(\mathbf{z})$
- Likelihood $p_{\theta}(\mathbf{x}|\mathbf{z})$
- Posterior $p_{\theta}(\mathbf{z}|\mathbf{x})$

1. First, sample a $\mathbf{z}^{(i)}$ from a prior distribution $p_{\theta^*}(\mathbf{z})$.
2. Then a value $\mathbf{x}^{(i)}$ is generated from a conditional distribution $p_{\theta^*}(\mathbf{x}|\mathbf{z} = \mathbf{z}^{(i)})$.

Variational AutoEncoder



- Prior $p_{\theta}(\mathbf{z})$
- Likelihood $p_{\theta}(\mathbf{x}|\mathbf{z})$
- Posterior $p_{\theta}(\mathbf{z}|\mathbf{x})$

ELBO:

$$\begin{aligned} \log p(S) \geq & - \sum_t \text{KL}[q_{\phi}(z_t|s_t) || p(z_t)] \\ & + \sum_t \mathbb{E}_{q_{\phi}(z_t|s_t)} [\log p_{\theta}(s_t|z_t)]. \end{aligned}$$

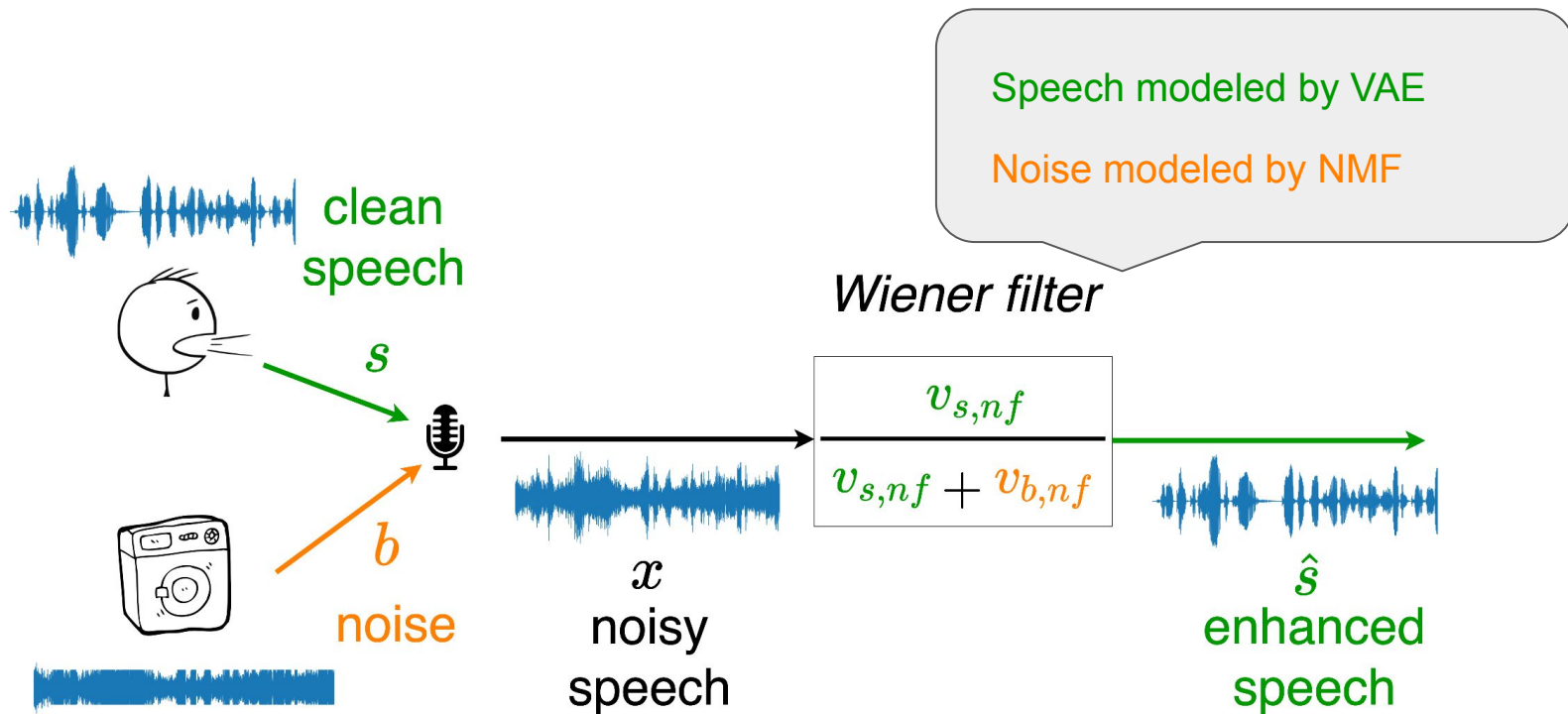
(Adapted from Bando et, al. Interspeech 2020)

VAE pros and cons

- + Flexible probabilistic graphical model
- Inexact likelihood calculation → blurry



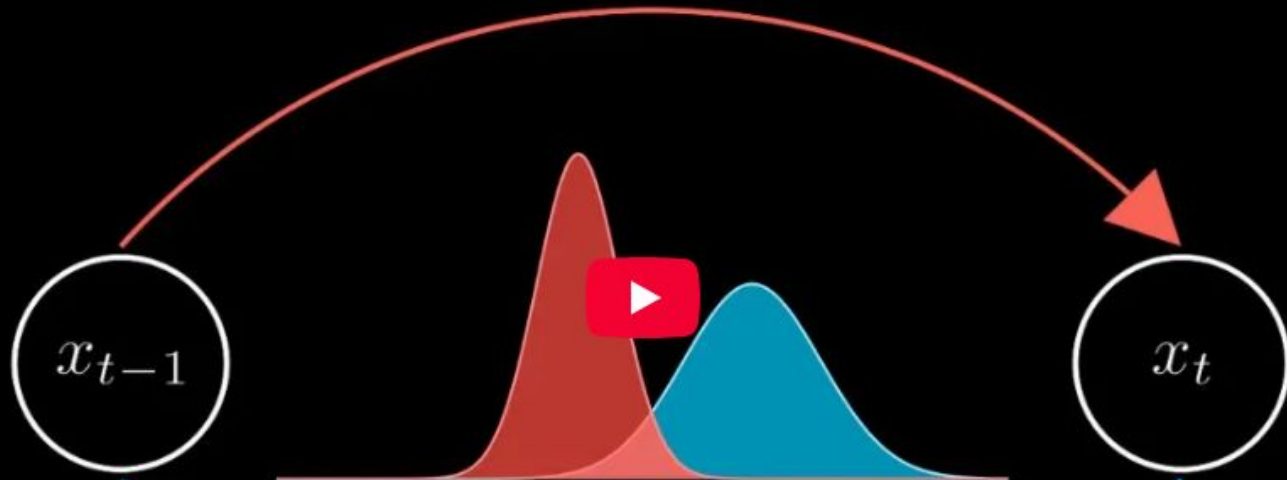
Speech Enhancement (hybrid)





Diffusion Models

$$q(x_t \mid x_{t-1})$$

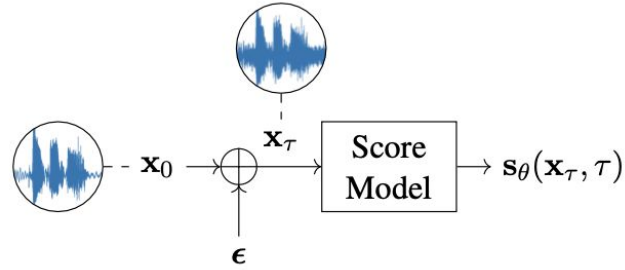


DDPM: <https://www.youtube.com/watch?v=EhndHhlyWWW>
Score-based Diffusion: <https://www.youtube.com/watch?v=IUljxdkoIK8>

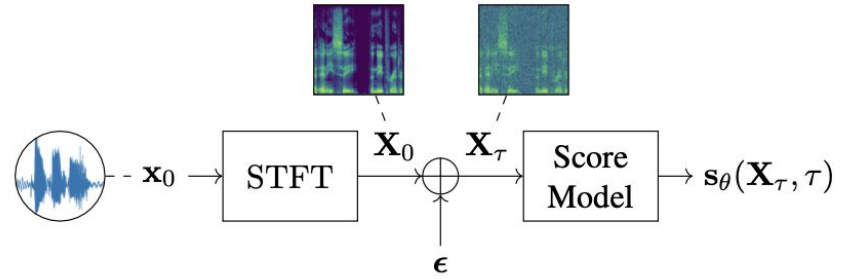
$$p_{\theta}(x_{t-1} \mid x_t)$$

Different diffusion models for SE

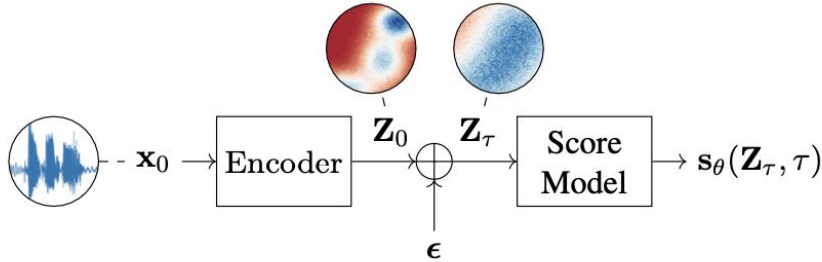
Waveform Diffusion



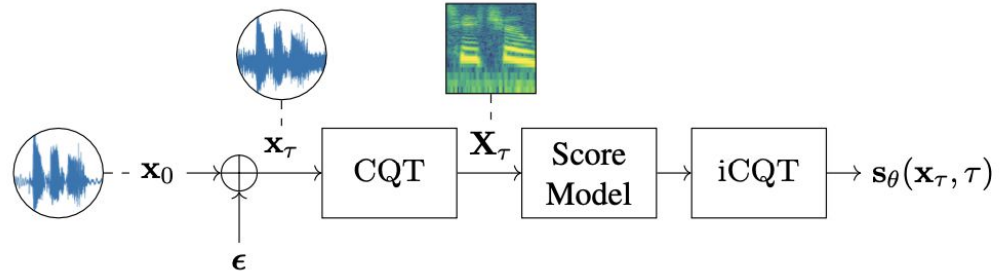
STFT Diffusion



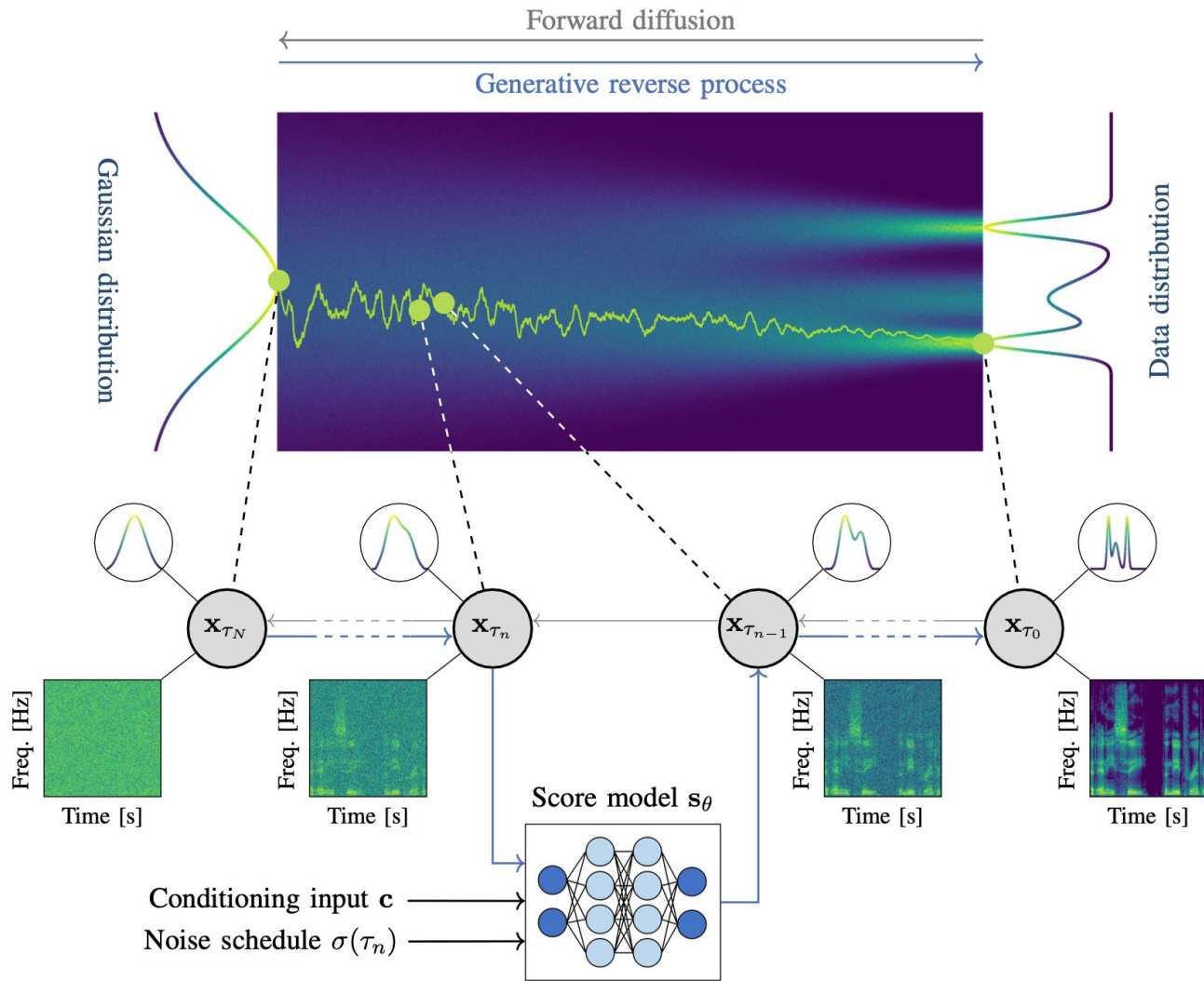
Latent Diffusion



CQT Score Estimation



Conditional Generation



J. -M. Lemerrier et al.,
 "Diffusion Models for Audio
 Restoration: A review," in IEEE
 Signal Processing Magazine,
 vol. 41, no. 6, pp. 72-84, Nov.
 2024

